



Cardiac neural representations for ECG-guided slice-to-volume reconstruction

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Abstract Cardiac 3D cine MRI is constrained by long acquisition times and challenging motion-handling requirements. As a result, certain patient conditions, such as cardiac arrhythmias, prevent the use of 3D sequences for whole-heart assessments. For these patients, 2D real-time sequences offer an alternative imaging solution, as they enable artifact-free acquisitions of individual slices with good in-plane contrast and temporal resolution. To derive whole-heart reconstructions from multi-planar 2D real-time MRI, we propose CiNeVol, a dynamic implicit neural representation network adapted from the static NeSVoR model. CiNeVol takes ECG- and respiration belt-derived motion states as inputs and provides decoupled estimates of cardiac and respiratory motion, yielding spatiotemporally consistent 3D reconstructions at arbitrary motion states. As light-weight in architecture, the model allows for fast instance optimization without requiring high-resolution training data. Evaluated on a simulated MRI dataset, CiNeVol demonstrates competitive image quality for both free-breathing (SSIM=0.92, PSNR=28.83, NMSE=0.03) and breath-hold (SSIM=0.89, PSNR=27.81, NMSE=0.04) 3D cine reconstructions, with faster runtimes than NeSVoR (SSIM=0.88, PSNR=26.13, NMSE=0.05) and SVRTK (SSIM=0.86, PSNR=26.28, NMSE=0.05). Preliminary in-vivo evaluations on three healthy volunteers confirm good spatiotemporal coherence, as reflected by superior expert scores, although remaining limitations, such as underestimation of cardiac contraction and image blurring, suggest the need for further optimization.

Keywords Cardiovascular MRI · Implicit neural representations · Motion correction · Slice-to-volume reconstruction · Super-resolution

1 Introduction

Cardiac 3D cine MRI is constrained by long acquisition times and challenging motion-handling requirements. There are two avenues of research that have been explored to tackle this issue: (i) self-gated, free-running 5D whole-heart magnetic resonance imaging (MRI) for reconstructing artifact-free cine images without requiring ECG triggers, breath-holding, or respiration navigators [1]; (ii) 3D volume reconstruction from “low-resolution” real-time 2D acquisitions [2]. For arrhythmic patients, electrocardiogram (ECG)

analysis remains a key aspect for the detection and classification of the origin of the pathological heartbeat, and 2D real-time sequences are the recommended solution for reliably capturing the cardiac contraction of individual heartbeats [2]. However, 2D sequences provide limited heart coverage, which is essential for accurate morphology and rhythm assessments. To derive motion-corrected volumetric cine MRI reconstructions from multiple (multi-planar) slice acquisitions, we explore slice-to-volume reconstruction (SVR) [3–5] techniques in image space. Various traditional SVR approaches [3, 4, 6] were proposed to recover volumetric information from multiple (anisotropic) slice observations while compensating for motion-induced structural inter-slice discontinuities and intensity variations. This involved solutions for intensity scaling, bias field estimation, edge-preserving regularization, and outlier removal. In the field of cardiac cine MRI, Odille et al. [7] proposed a two-stage SVR approach for multi-planar, balanced steady-state free precession (bSSFP) MRI, first deriving

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motion-corrected 3D+t reconstructions using cine-GRICS [8] with ECG-guidance, then solving for a super-resolved isotropic volume combining the corrected stacks. Amerom et al. [5] contributed to the SVRTK framework [9], proposing a multi-stage pipeline for interleaved rigid motion correction and super-resolution reconstruction of fetal cardiac cine MRI using image-based cardiac phase estimation for cardiac synchronization. Although deformable SVR variants were proposed to model non-rigid motion by multilevel B-splines [9], their high computational complexity impedes their application to dynamic data. Supervised deep learning-based solutions [10, 11] enable fast reconstructions at inference time but require large training datasets with paired low-resolution and high-resolution MRI to learn generalizable models. Unsupervised networks, such as implicit neural representations, overcome this requirement by solely considering low-resolution images [12, 13]. Originally proposed for computer vision applications [14], implicit neural representation (INRs) have gained increasing interest in the field of medical imaging. Using a lightweight, self-supervised, instance optimization paradigm, neural representations overcome large GPU requirements and generalization challenges typically encountered by deep learning-based supervised methods. They learn continuous functions that map input coordinates to physical properties of the scene (e.g. MR intensity at (x, y, z)), which results in efficient, resolution-agnostic representations that support arbitrary inference voxel grids and resolutions. One of the first implicit SVR methods for MRI reconstruction was proposed by Wu et al. [15], who processed multi-planar, motion-corrected static brain images. Yeung et al. [16] proposed an INR for 3D ultrasound reconstruction using a separate, pre-trained model for automated slice localization. More recently, Xu et al. [17] presented NeSVoR, a fast INR network for static SVR of fetal brain MRI, incorporating concepts such as multi-resolution hash encodings for computational speed-up [18], learnable bias fields, and reconstruction uncertainty maps. In the field of computer vision, promising progress was also achieved for the modeling of moving scenes [19, 20] and the efficient encoding of spatiotemporal features [21].

Aiming to reconstruct spatiotemporally consistent 3D cine MRI from free-breathing 2D real-time cine acquisitions, we present CiNeVol (Cine Neural Volume), an implicit neural representation model for dynamic slice-to-volume reconstruction. Guided by motion states derived from ECG and respiration belts, the model enables decoupled learning of cardiac and respiratory motion fields, transforming all moving 2D cine observations into a canonical reference space. In this reference space, super-resolved volume reconstructions can be generated for queried motion states. For preliminary validation, we evaluate the model on simulated

3D cine MRI and in-vivo real-time MRI of three healthy volunteers.

2 Methods

2.1 The CiNeVol model

2.1.1 Problem formulation

In static SVR, we solve for a high-resolution 3D volume V , consisting of voxels $\{v_n\}_{n=1}^N$, by combining K observed (anisotropic) 2D slices $\{Y_k\}_{k=1}^K$ with associated pixel intensities $\{i_{jk}\}_{j=1}^J$ [4]:

$$I_k^* = M_k V; \quad i_{jk}^* = s_k \exp(-b_{jk}) i_{jk}. \quad (1)$$

The row m_{nj}^k of matrix M_k describes the acquisition of pixel i_{jk} from the volume V and comprises a spatially aligned point spread function (PSF) that relates the volume to the observed pixel. Opposed to static scenes, for which the PSF can be defined based on the known slice resolution and position, moving scenes require the incorporation of deformation models that estimate the unknown spatial relationship between i_{jk} and V . A changing sample position in the scanner furthermore impacts the pixel intensities, often compensated by slice-dependent intensity scales s_k and bias fields b_{jk} . Some approaches furthermore model outlier weights p_{jk} to limit the impact of artifacts or poorly aligned samples and add a (edge-preserving) regularization term R for stability, solving the following inverse problem [4]:

$$V = \underset{V}{\operatorname{argmin}} \sum_j \sum_k p_{jk} \|i_{jk}^* - \sum_n m_{nj}^k v_n\|^2 + \lambda R(V). \quad (2)$$

Interleaved iterations of registration optimization (update of M_k) and super-resolution reconstruction are typically applied using gradient descent. In dynamic cine SVR, this formulation is extended by a temporal dimension that represents the cardiac phase, considering 2D+t slice acquisitions I_{kt} and a 3D+t target volume [5]. In our work, we consider the explicit modelling of a further temporal dimension to model the respiration motion state, resulting in 5D target volumes V .

2.1.2 Proposed INR (CiNeVol)

The proposed CiNeVol network, an adaptation of NeSVoR [17] for dynamic SVR of cardiac cine MRI, is visualized in Fig. 1. In contrast to the NeSVoR model [17], which learns static 3D representations and applies rigid slice-to-volume transformations to correct inter-slice motion,

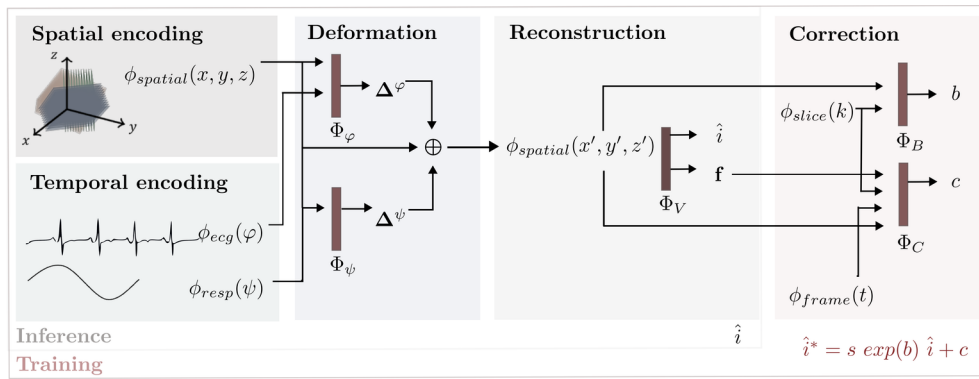


Fig. 1 The proposed CiNeVol architecture. Network inputs are the spatial coordinates (x, y, z) , the cardiac motion states φ , and the respiratory motion states ψ . Inputs are hash encoded using the decomposed grids $\phi_{spatial}$, ϕ_{ecg} , and ϕ_{resp} , and the resulting encodings are passed to two deformation multi-layer perceptrons (MLPs), visualized as red bars, that predict motion state-dependent deformation offsets $(\Delta_\varphi, \Delta_\psi)$, mapping each spatiotemporal pixel input location to a 3D reference space. In this reference space, intensity predictions $\hat{i}$ are derived by the MLP Φ_V and, at training time, corrected by estimated

intensity scales s , bias fields b , and correction values c . The bias field is predicted by the MLP network Φ_B conditioned on the embedded slice index $\phi_{slice}(k)$. The MLP network Φ_C estimates a correction field, and is conditioned on the embedded slice index $\phi_{slice}(k)$, the embedded frame index $\phi_{frame}(t)$, and an intermediate feature vector f . At inference time, 5D intensity reconstructions can be queried for dense inference grids and arbitrary motion states, while the correction operations are discarded

CiNeVol models a 5D spatiotemporal feature space and integrates cardiac and respiratory motion fields. These motion fields are represented through additional multi-layer perceptron (MLP) networks that receive motion states derived from physiological sensor data, enabling the model to learn subject-specific dynamic deformation patterns. The pipeline is optimized per subject using a dataset of K (multi-planar, anisotropic) 2D+t slice acquisitions with T data frames I_{kt} . As input, the network receives the spatial pixel locations $\mathbf{x}_{jk} := (x_{jk}, y_{jk}, z_{jk})$ and the cardiac and respiratory motion states φ_{tk} and ψ_{tk} (see Sec. 2.2.3). Learnable multi-resolution hash encodings [18] map these input coordinates into a higher-dimensional feature space, allowing efficient learning of both fine and coarse spatial structures with rapid convergence. As 5D hash grids are inefficient for sparse temporal dimensions and prone to hash collisions, we employ a decomposed hash encoding [21], consisting of seven individual 3D grids to encode the 5D feature space. This includes a purely spatial encoding $\phi_{spatial}$ for (x, y, z) , a spatiotemporal cardiac motion state encoding ϕ_{ecg} (concatenating the encodings of (x, y, φ_{tk}) , (x, z, φ_{tk}) , and (y, z, φ_{tk})), and a spatiotemporal respiration state encoding ϕ_{resp} (concatenating the encodings of (x, y, ψ_{tk}) , (x, z, ψ_{tk}) , and (y, z, ψ_{tk})). We decouple the dynamic reconstruction task into two subtasks: the estimation of deformation offsets Δ to transform pixels of all frames to a canonical reference space, and the prediction of intensity reconstructions in this 3D space. Aiming for disentangled representations of cardiac and respiratory motion, we optimize two independent MLPs, Φ_φ and Φ_ψ , for the deformation estimation, with:

$$\Delta_{jkt}^\varphi = \Phi_\varphi(\phi_{spatial}(\mathbf{x}_{jk}), \phi_{ecg}(\varphi_{tk})); \quad (3)$$

$$\Delta_{jkt}^\psi = \Phi_\psi(\phi_{spatial}(\mathbf{x}_{jk}), \phi_{resp}(\psi_{tk})). \quad (4)$$

We then derive the updated pixel location, describing the deformed object in the canonical space, by combining the two offsets:

$$\mathbf{x}'_{jkt} = \mathbf{x}_{jk} + \Delta_{jkt}^\varphi + \Delta_{jkt}^\psi. \quad (5)$$

At the predicted pixel location, a new spatial feature encoding $\phi_{spatial}(\mathbf{x}'_{jkt})$ is derived and passed to the intensity MLP Φ_V , which predicts an intensity value $\hat{i}$, an intermediate feature vector f , and—when segmentation supervision is enabled—cardiac segmentation mask estimates $\mathbf{o}$ with class probabilities corresponding to background, left ventricle (LV), myocardium, and right ventricle.

$$\hat{i}_{jkt}, f_{jkt}, \mathbf{o}_{jkt} = \Phi_V(\phi_{spatial}(\mathbf{x}'_{jkt})). \quad (6)$$

To compensate for intensity variations between individual frame observations, we follow [17] and optimize K intensity scale parameters s_k for slice-wise intensity adaptation and use two further networks Φ_B and Φ_C to predict a bias field and a correction value, respectively. Identical to NeSVoR, the bias network Φ_B is conditioned on the encoded slice index $\phi_{slice}(k)$ and receives hash features of the coarsest L_b resolution levels to capture low-frequency variations:

$$b_{jkt} = \Phi_B(\phi_{spatial}(\mathbf{x}'_{jkt})[0 : L_b F], \phi_{slice}(k)). \quad (7)$$

We extend the correction MLP Φ_C to be conditioned not only on the encoded slice indices $\phi_{slice}(k)$ and the intermediate feature vector $\mathbf{f}$, as in NeSVoR, but also on the encoded frame indices $\phi_{frame}(t)$, thereby enabling Φ_C to capture temporal variations:

$$c_{jkt} = \Phi_C(\phi_{spatial}(\mathbf{x}'_{jkt}), \mathbf{f}_{jkt}, \phi_{slice}(k), \phi_{frame}(t)). \quad (8)$$

As in [17], we derive corrected intensity predictions by combining the slice scale, bias field, and intensity predictions:

$$\hat{i}_{jkt}^* = s_k \exp(b_{jkt}) \hat{i}_{jkt} + c_{jkt}, \quad (9)$$

and approximate the spatial sinc PSF by an anisotropic 3D Gaussian with full width at half maximum (FWHM) set to the slice-thickness in the through-plane direction and to $1.2 \times$ voxel size in the in-plane directions. PSF blurring is then introduced by averaging network predictions Eq. 9 obtained for U randomly drawn pixel locations $\{\mathbf{x}_{jktu}\}_{u=1}^U$ from the slice orientation-aligned Gaussian distribution [17] (Fig. 2).

2.1.3 Objective functions

The predictions $\hat{i}_{jkt}^*$ are evaluated against the target pixel intensity observations i_{jkt} using the mean absolute error (MAE) as reconstruction loss, where ω represents a random batch sample index jkt and Ω the batch size. We add a total variation (TV) term for intensity regularization, exploiting PSF samples to model a local neighborhood (comparing pairs of random samples, with $l = u + U/2$) as in [17]:

$$L_{MAE} = \frac{1}{\Omega} \sum_{\omega=1}^{\Omega} |\hat{i}_{\omega}^* - i_{\omega}| \text{ and } L_{TV} = \frac{1}{\Omega U} \sum_{\omega=1}^{\Omega} \sum_{u=1}^{U/2} \frac{|\hat{i}_{\omega u} - \hat{i}_{\omega l}|}{\|\mathbf{x}'_{\omega u} - \mathbf{x}'_{\omega l}\|}. \quad (10)$$

When segmentation supervision is enabled, the predicted segmentation masks are optimized using the cross-entropy

loss L_{Seg} . To encourage the network to maximize the information contained in the joint intensity predictions $\hat{i}_{jkt}$, we assign a penalty on the correction field estimations $\hat{c}_{jkt}$, as in NeSVoR:

$$L_{Corr} = \frac{1}{\Omega} \sum_{\omega=1}^{\Omega} |c_{\omega}|. \quad (11)$$

As [17] we optimize the bias field to be close to zero but add additional smoothness regularization.

$$L_{Bias} = \left(\frac{1}{\Omega} \sum_{\omega=1}^{\Omega} |b_{\omega}| \right)^2 + \frac{1}{\Omega U} \sum_{\omega=1}^{\Omega} \sum_{u=1}^{U/2} \frac{|b_{\omega u} - b_{\omega l}|}{\|\mathbf{x}'_{\omega u} - \mathbf{x}'_{\omega l}\|} \quad (12)$$

To regularize the estimated deformation fields, we employ Jacobian regularization for both deformation MLP network outputs [22], with Jacobian matrix $J \in \mathbb{R}^{3 \times 3}$, identity matrix $\mathbb{I} \in \mathbb{R}^{3 \times 3}$, and Frobenius norm F :

$$L_{Jac} = \frac{1}{\Omega} \sum_{\omega=1}^{\Omega} (\|J_{\omega}^T J_{\omega} - \mathbb{I}\|_F)^2, \quad (13)$$

In segmentation-supervised settings, we further penalize cardiac deformation estimates extending beyond the cardiac region:

$$\mathcal{L}_{reg} = \frac{1}{\Omega} \sum_{\omega=1}^{\Omega} \tilde{d}_{\omega} \|\Delta_{\omega}^{\varphi}\|_2^2, \quad (14)$$

where $\Delta_{\omega}^{\varphi}$ denotes the predicted deformation vector at voxel ω , and $\tilde{d}_{\omega}$ represents the normalized squared distance from the estimated heart center, with $\tilde{d}_{\omega} = 0$ for voxels inside the heart mask.

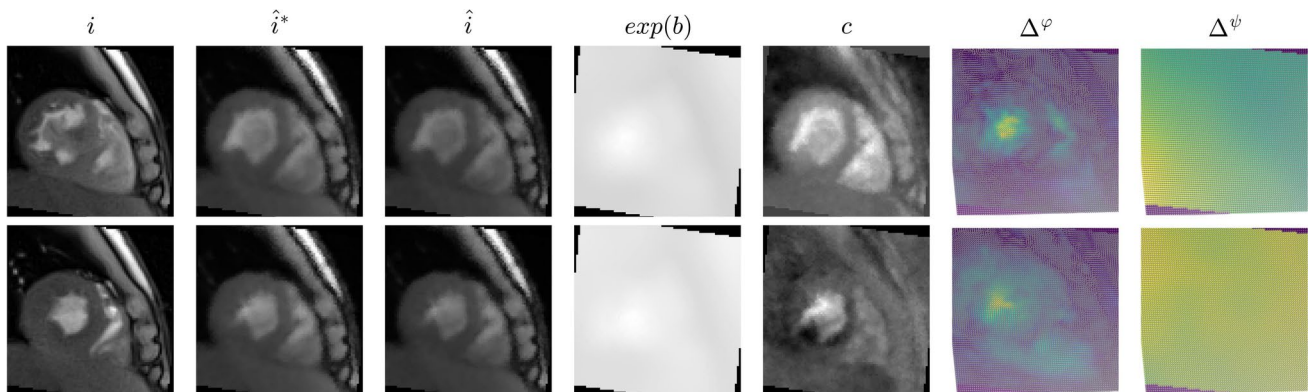


Fig. 2 Intermediate Proposed+seg model predictions for a healthy volunteer. From left to right: target, corrected intensity prediction, raw intensity prediction, bias field, correction field, and deformation fields,

where the color represents the motion vector magnitude and the vector orientation indicates the motion direction in 3D space

Combining these loss contributions, the final loss function is defined as:

$$L = L_{MAE} + \lambda_{TV} L_{TV} + \lambda_{Seg} L_{Seg} + \lambda_{Bias} L_{Bias} + \lambda_{Corr} L_{Corr} + \lambda_{Jac}^{\varphi} L_{Jac}^{\varphi} + \lambda_{Jac}^{\psi} L_{Jac}^{\psi} + \lambda_{Reg} L_{Reg}. \quad (15)$$

At inference, we sample PSF samples using an isotropic Gaussian and discard the networks Φ_B and Φ_C , providing $\hat{i}$ as the final reconstruction.

2.2 Datasets

2.2.1 Simulated dataset

To enable method evaluation under controlled deformation fields and reference reconstructions, we explored the XCAT phantom software [23] and the MRXCAT2.0 [24, 25] MRI simulation software for the generation of five phantom datasets. Using XCAT, we derived 3D+t anatomical label maps and deformation fields for free-breathing body movement simulations. The software parameter files were adjusted to generate one female and four male phantoms at a spatial resolution of 1.0 mm^3 and a temporal resolution of 50 ms (size $300 \times 300 \times 250$). The simulated acquisitions covered 1–2 respiration and 5–7 cardiac cycles, with cycle durations of 3.33–5.0 and 0.86–1 seconds, respectively. Two phantoms were generated with simulated myocardial lesions: one exhibiting a motion defect (reduced contraction) and the other maintaining normal contraction like the surrounding myocardium. To convert the 3D+t XCAT phantoms to texturized MRI simulations, we employed the 2D texture generation network of [25], yielding texturized PD, T1, and T2 maps for each image frame. The individual frame predictions were then stacked to 3D+t tissue maps and passed to the MR simulation pipeline of [24], which yielded bSSFP MR-like images (TR/TE: 3.0/1.5ms, flip angle: 60° , SNR: 20, coils: 8, low-pass filter strength: 3.0). Once the texturized 3D+t phantoms were generated in reference orientation, we simulated anisotropic short axis (SAX), vertical long axis (VLAX), and horizontal long axis (HLAX) image stacks by first resampling the 3D+t image to the desired orientations and resolution ($1.48 \times 1.48 \times 8 \text{ mm}$) and then extracting one cardiac cycle of a random respiration period per slice to induce breathing motion-related inter-slice discontinuities. This resulted in three imaging stacks per phantom, each exhibiting 30 slices and 17–20 temporal frames. For free-breathing evaluation, we resampled a sub-sequence of the 3D+t reference view volume into a SAX view with 1.48 mm^3 isotropic resolution, covering one cardiac cycle during inhale motion (*Phantom (FB)*). To enable the model performance comparison to related SVR approaches that compensate for respiration motion, we furthermore created

breath-hold reference phantoms for two of the five phantom configurations (covering one cardiac cycle in a fixed exhale breathing motion state) (*Phantom (BH)*).

2.2.2 In-vivo dataset

Three healthy subjects were scanned on a 3 T Siemens scanner using a 2D+t real-time FLASH gradient echo sequence with interleaved radial sampling [2] at a spatial resolution of $1.48 \times 1.48 \times 6 - 8 \text{ mm}$ and a temporal resolution of 20 ms (TR/TE: 2.22–2.42.22.42/1.46ms, flip angle: 8° , bandwidth: 1470 Hz, FOV: $296 \times 296 \text{ mm}$). The 2D+t cine data was acquired consecutively in SAX, VLAX, and HLAX view planes, resulting in anisotropic imaging stacks with each 8–21 slices and 65–80 temporal frames. All subjects were breathing freely during the acquisition. The data was acquired under the approved ethical protocol “METHODODO” (ClinicalTrials.gov Identifier: NCT02887053). The study was approved by the institutional ethics review board (CPP EST-III, 08.10.01). All participants provided informed written consent. The ECG was captured using a prototype from the Maglife system (Schiller Médical, Wissembourg, France) at 16 kHz [26]. The signal analyzer and event controller (SAEC) [27] was employed for ECG-based triggering allowing the slice to be changed at the R-wave detection. The number of pre-scans required for the NLInv algorithm [2] was individually adjusted for each subject to cover approximately 75% of a cardiac cycle, and the number of measurements was adjusted to the duration of approximately 125% of a cycle. This ensured that at least one complete cardiac cycle per slice was acquired. The ECG data was recorded using the SAEC system. Additionally, one of the three subjects was monitored using a respiration belt (Maglife, Schiller Médical, Wissembourg, France).

2.2.3 Motion phase definitions

We used a bilinear cardiac phase definition as proposed by Feinstein et al. [28] ($\varphi_t \in [0, 1]$) using the R-peak detection of the SAEC system to define QRS complex locations and assigning motion states for systolic ($(t - t_{QRS}) < \Delta_{sys}$) and diastolic time points t as:

$$\varphi_{t_{systole}} = \frac{t - t_{QRS}}{\Delta_{sys}} \frac{\overline{\Delta_{sys}}}{\overline{\Delta_{RR}}} \quad \text{and} \quad \varphi_{t_{diastole}} = \frac{t - t_{QRS} - \Delta_{sys}}{\Delta_{dia}} \frac{\overline{\Delta_{dia}}}{\overline{\Delta_{RR}}} + \frac{\overline{\Delta_{sys}}}{\overline{\Delta_{RR}}} \quad (16)$$

with Δ_{RR} denoting the current RR-interval duration, t_{QRS} the preceding QRS location, Δ_{sys} the systolic phase duration, $\Delta_{dia} = \Delta_{RR} - \Delta_{sys}$, $\Delta_{sys} = (546 \text{ ms} - 2.1 \text{ HR}_t)/1000$, $\text{HR}_t = 60/\Delta t_{RR}$, and $\bar{x}$ the full record mean values. For

the subject for which respiration recordings were available, the signal was low- (3Hz) and high-pass (0.1Hz) filtered using a Butterworth filter. For the remaining subjects we derived respiration estimations from the ECG signals, by the subsequent application of a Butterworth high-pass filter (0.15 Hz) to compensate for signal drift and a low-pass filter (0.25–0.5 Hz) to extract the low-frequency respiration curve. After scaling both the acquired and predicted respiration signals to the range $[0, 1]$, the respiration motion states were determined based on the signal amplitude.

2.3 Implementation details

Implementation details for all model setups can be found in the supplemental material. For the CiNeVol setup, we compared training with (Proposed+seg) and without (Proposed) segmentation mask supervision. In the segmentation-supervised setup, we derived SAX input stack segmentation masks using a 2D nnU-Net model trained on the M&Ms challenge dataset [29]. The baseline INR consisted solely of Φ_V , Φ_B , and Φ_C . In this baseline configuration, no explicit deformation field estimation was performed, instead, the encoded 5D inputs $\phi_{spatial}(x, y, z)$, $\phi_{ecg}(\varphi)$, and $\phi_{resp}(\psi)$ were directly passed to the intensity MLP Φ_V , learning a direct mapping from spatiotemporal space to the intensity reconstruction. For comparison, we additionally considered Lanczos-windowed sinc interpolation, SVRTK-based 4D cardiac reconstruction [5, 9], and NeSVoR [17], training one model per cardiac phase and stacking the individual predictions into 4D reconstructions. In the Interpolation setup, we considered 17–20 cardiac phases, averaged the temporal frames of each cardiac phase for the SAX stack, and interpolated the result to isotropic resolution using Lanczos interpolation. Opposed to the proposed INRs, which enabled querying desired respiration motion states at inference, SVRTK and NeSVoR compensate for respiration motion without explicit respiration state control. To ensure approximate alignment across the individual 4D reconstructions, we performed rigid registration (ANTs [30]), considering the temporal average volumes to optimize static rotation and translation parameters. Registration targets were the reference volumes for the phantom data and the interpolated volumes for the real-time data. As the ROIs varied across approaches, we masked all registered 4D breath-hold reconstructions by the aligned cardiac SVRTK mask and evaluated quality metrics in the mask-enclosing bounding box.

2.4 Method evaluation

2.4.1 Evaluation metrics

As reference metrics, we considered the structural similarity (SSIM), the peak signal-to-noise ratio (PSNR),

and the root-mean-square error (RMSE). Left ventricle volumes were estimated from (uncorrected) predictions of a 3D nnU-Net model trained on the M&Ms challenge dataset [29]. Stroke volume (SV) was derived as the difference between the end-diastolic frame ($t = 0$) and the end-systolic frame (identified as the minimum of the measured LV volume curve). As no-reference metric, we assessed the sharpness index (SI) [31] to reflect image detail loss caused by blurring and noise (calculated in-plane SI_{in} and through-plane $SI_{through}$). While SSIM, PSNR, and NMSE were computed per 3D volume and averaged across time points, the SI score was calculated per 2D frame, and then averaged across slices and time points. Finally, an expert with 5 years of experience in cardiac MRI analysis assessed image quality, blinded to the method, using a 5-point scoring system [5]: 4) high-image quality and distinct appearance of cardiac structures, 3) adequate image quality to determine most details, 2) sufficient image quality to determine some details, 1) poor image quality with significant lack of detail, and 0) inadequate image quality to visualize any cardiac structure.

3 Results

3.1 Simulated dataset

The results of the phantom experiments are summarized in Tables 1 to 2. Note that while the *Phantom (FB)* and *Phantom (BH)* setups share the same type of motion-corrupted training data, they differ in the motion states queried at inference, the evaluation targets used, and the number of samples included. Since the presented methods (Baseline, Proposed, and Proposed+seg) support querying respiratory motion states at inference, whereas SVRTK, NeSVoR, and Interpolation do not, the breath-hold reference results (Table 2) represent the fairest comparison of reference quality metrics. Fig. 3 demonstrates example free-breathing 3D cine reconstructions for the Baseline and Proposed model for two phantoms with simulated myocardial abnormalities. The motion profiles of the Proposed model indicate accurate modeling of both respiratory and cardiac motion, as well as motion defects. Overall, the Proposed INR produced more consistent FB representations than the simple mapping of the Baseline INR, whose predictions were comparatively noisy, with SSIM, PSNR, and NMSE scores of 0.92, 29.26, 0.03, and 0.86, 27.06, 0.04, respectively. In the FB setup, the CiNeVol model achieved LV volumes closest to the reference, improving further from 1.60 (Proposed) to 1.05 (Proposed+seg) when incorporating

Table 1 Performance comparison for the free-breathing phantom dataset (*Phantom (FB)*, N=5)

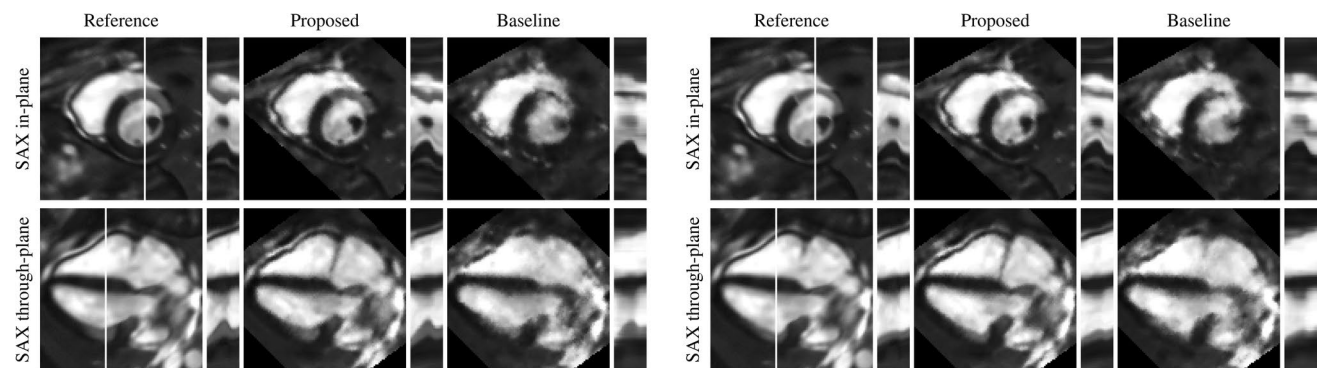
Method	Quality				Sharpness		Runtime Time (s)
	SSIM $\uparrow$	PSNR $\uparrow$	NMSE $\downarrow$	SV diff. $\downarrow$	SI_{in} $\uparrow$	$SI_{through}$ $\uparrow$	
Proposed+seg	0.92	28.83	0.03	1.05	31.44	30.46	504.20
Proposed	0.92	29.26	0.03	1.60	31.32	30.39	503.40
Baseline	0.86	27.06	0.04	15.52	28.64	27.35	242.80
NeSVoR	0.81	24.66	0.08	2.40	35.25	31.48	2735.80
SVRTK	0.83	25.34	0.07	2.86	33.75	32.43	1334.80
Interpolation	0.37	18.67	0.28	8.47	27.50	26.63	–

Best mean performance metrics are shaded gray. Scores for NeSVoR, SVRTK, and Interpolation are shown for completeness but are not directly comparable, as these methods do not support querying respiratory motion states at inference time

Table 2 Performance comparison for the breath-hold phantom dataset (*Phantom (BH)*, N=2)

Method	Quality				Sharpness		Runtime Time (s)
	SSIM $\uparrow$	PSNR $\uparrow$	NMSE $\downarrow$	SV diff. $\downarrow$	SI_{in} $\uparrow$	$SI_{through}$ $\uparrow$	
Proposed+seg	0.89	27.81	0.04	10.71	30.14	28.76	501.50
Proposed	0.89	27.87	0.03	9.71	30.28	28.48	500.50
Baseline	0.81	25.03	0.07	11.09	27.48	25.33	241.50
NeSVoR	0.88	26.13	0.05	2.92	36.71	31.51	2838.50
SVRTK	0.86	26.28	0.05	0.52	32.88	31.49	1321.00
Interpolation	0.35	18.43	0.29	4.40	26.35	24.60	–

Best mean performance metrics are shaded gray

**Fig. 3** Free-breathing phantom reconstructions of a phantom without motion defect (left) and with motion defect (right), shown in SAX and through-plane views. The time profiles show the cardiac contraction over the entire cardiac cycle for a pixel column (highlighted in white)

the segmentation task. In the breath-hold inference setup (*Phantom (BH)*), the lowest image quality was observed for the Interpolation approach (SSIM = 0.35, PSNR = 18.43) and the comparatively fast Baseline INR (SSIM = 0.81, PSNR = 25.03), followed by SVRTK (SSIM = 0.86, PSNR = 26.28). NeSVoR (SSIM = 0.88, PSNR = 26.13) and the proposed methods, Proposed (SSIM = 0.89, PSNR = 27.87) and Proposed+seg (SSIM = 0.89, PSNR = 27.81), achieved very similar qualitative performance metrics. With an average computation time of about 500 seconds, the proposed INR required less than half of the optimization time needed by SVRTK. Note that the reported runtimes for the NeSVoR model

represent accumulated runtimes across cardiac phases. As the models were trained independently, acceleration could be achieved through parallelization. Compared to SVRTK and NeSVoR, larger stroke volume estimation discrepancies were observed for both Proposed and Proposed+seg, yielding greater reference SV differences. Image sharpness was highest for SVRTK and NeSVoR across all experiments. Example breath-hold inference reconstructions are visualized in Fig. 4, demonstrating that all SVR models successfully compensated for the inter-slice discontinuities present in the Lanczos resampling reconstruction, while showing differences in cardiac contraction profiles and image contrast.

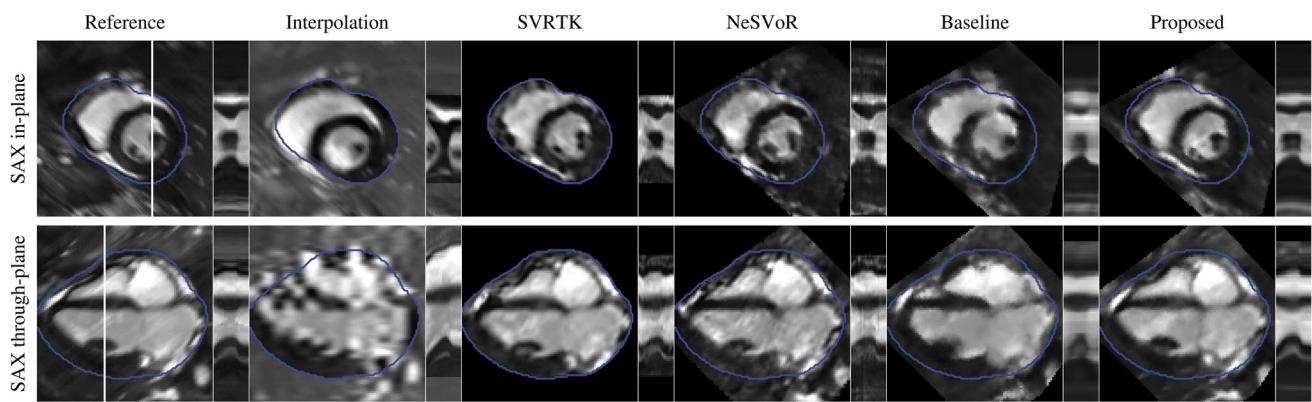


Fig. 4 Rigidly registered breath-hold phantom reconstructions in SAX view with evaluation mask contour overlay in blue. The time profiles show the cardiac contraction for a pixel column (highlighted in white)

3.2 In-vivo dataset

An example reconstruction for a healthy volunteer, for which both ECG and respiration recordings were available, is shown in Fig. 5. The proposed INR yielded visually most convincing spatiotemporal consistency and inter-slice shift motion correction, as confirmed by the expert rating. However, it produced comparably smooth predictions, resulting in inferior image sharpness metrics, as reported in Table 3. Despite promising motion profiles, we observed

an underestimation of the cardiac contraction during the systolic phase (Fig. 6). Adding a segmentation task to the INR pipeline (Proposed+seg) and penalizing deformation field estimations outside the cardiac region aimed to improve the estimation of cardiac contraction. This resulted in a slight improvement in LV volume estimation during the systolic phase but still produced lower stroke volume estimates compared to the NeSVoR model, which was trained separately for each cardiac phase. Artifacts of the SVRTK approach resulted in low expert scores and

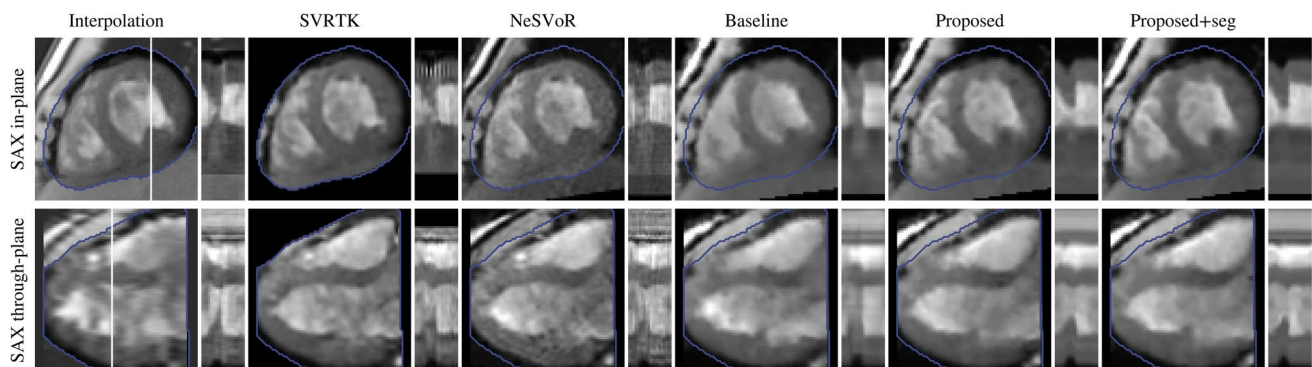


Fig. 5 Example reconstructions for a real-time MRI volunteer in SAX view with evaluation mask contour overlay in blue. The time profiles show the cardiac contraction for a pixel column (highlighted in white)

Table 3 Performance comparison for the in-vivo dataset (N=3)

Method	Expert scoring	Sharpness		Stroke volume SV estim.	Runtime Time (s)
		$0 - 4 \uparrow$	$SI_{in} \uparrow$ $SI_{through} \uparrow$		
Proposed+seg	2.67	13.32	13.08	48.99	622.67
Proposed	2.67	13.62	13.27	41.52	619.67
Baseline	1.33	13.45	13.32	53.07	302.00
NeSVoR	1.67	19.85	17.15	62.18	2923.33
SVRTK	0.67	20.61	17.73	38.79	2867.67
Interpolation	—	17.19	14.39	61.75	—

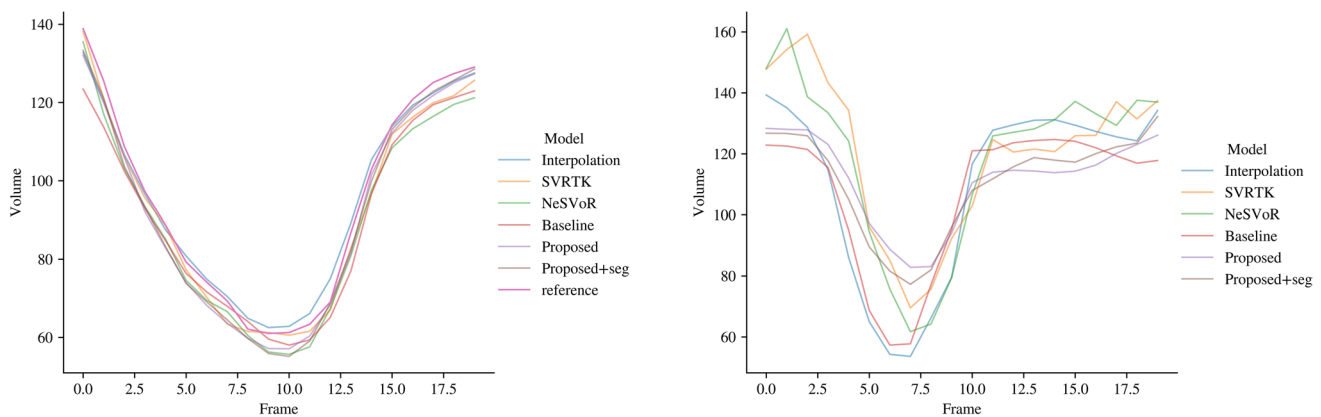


Fig. 6 Left ventricle volume estimations for a free-breathing phantom (left) and an in-vivo subject (right)

inaccurate nnU-Net model predictions for two subjects, leading to unreliable stroke volume estimations for the in-vivo dataset. Reconstructions for all in-vivo subjects are presented in the supplemental material.

4 Discussion

Major strengths of the proposed CiNeVol model include its comparatively fast runtime and its ability to jointly model the spatial and temporal domains, yielding coherent free-breathing reconstructions. Enforcing all samples to be mapped to a canonical reference space via deformation fields imposed strict consistency constraints, which may have limited the model's ability to capture fine textural details, resulting in comparatively smooth reconstructions. The observed performance decrease of the Baseline, Proposed, and Proposed+seg models when comparing free-breathing to breath-hold inference phantom setups may be due to insufficient coverage of the 5D training data space, as well as correlated cardiac and respiratory motion states in the simulated training data. Such correlation could have caused entangled motion representations, limiting the network's ability to capture the heart cycle at maximal exhale. Underestimation of systolic contraction was even more pronounced in the in-vivo dataset. Incorporating segmentation masks as anatomical priors appeared to improve motion estimation, however, additional strategies are needed to achieve accurate patterns of cardiac contraction. Potential approaches include advanced regularization to prevent the model from encoding motion-related changes in the correction fields (Fig. 2) and refined slice- and frame encodings (e.g., using a contrastive loss to encourage similar representations for nearby motion states [32]). Given the relatively short overall MRI acquisition time, gross patient motion is expected to be minimal. Nevertheless, future extensions of the model could incorporate strategies to compensate for

general patient motion, either by suppressing corrupted slices or by learning additional motion correction modules. These modules could rely on learnable rigid motion parameters [17] or model an additional motion field. Regarding the methodological comparison, it is worth noting that the SVRTK-based SVR pipeline was originally designed for fetal imaging, consequently, our data setup may have violated certain method assumptions, such as sufficient spatial and temporal coverage and ROI definitions. The NeSVoR model was originally designed for static reconstructions, but in this work, it was applied to assess cardiac motion using stacked 4D predictions. The presented simulated phantom data had certain limitations that could be further improved. Notably, the data generation relied on a pre-trained 2D texture U-Net applied to generate 3D+t phantoms without additional intra-slice corrections. Compared to real MRI scans, the simulated data exhibited fewer intensity variations caused by bias field inhomogeneities, artifacts, or patient motion (particularly blood flow). Additionally, by using ground-truth motion states for training, the impact of motion measurement noise was not assessed. Performance differences between the three human volunteers could, among others, be attributed to imperfect respiration signal predictions for subjects without respiration belt recordings and varying slice thicknesses.

This study was motivated by the need for flexible imaging techniques for arrhythmia patients, but, at this stage, did not include any in-vivo arrhythmia patients. Since the proposed model is conditioned on ECG-based motion states, additional adaptations incorporating advanced ECG analysis would be needed to provide appropriate motion encodings for irregular rhythm disorders. A strength of our data acquisition setup is that the real-time sequence enables on-the-fly reconstructions of both normal and abnormal beat sequences. When abnormal beats are detected, they can either be omitted to reconstruct the normal rhythm exclusively, or incorporated by assigning suitable cardiac motion

states to represent both normal and abnormal sequences. As opposed to supervised models that often fail to generalize to unseen anatomical features, INRs can capture instance-specific properties (such as the myocardial tissue abnormalities and regular motion defects presented in Fig. 3). A challenge is that pathological beats often occur less frequently than normal contractions, introducing a risk of bias toward normal beat patterns. Promising research directions to improve model robustness under few-shot learning scenarios include meta-learning and hyper-network approaches [33], as well as the incorporation of patient-specific shape or motion priors, for example through the integration of population-trained image encoders [34]. These strategies can help regularize the prediction space, enabling fine-tuning focused specifically on patient- or arrhythmia-specific details, which could further improve runtimes. Additional speedups could be achieved through adaptive sampling strategies that concentrate computation on anatomically relevant regions, as well as through coarse-to-fine reconstruction pipelines.

5 Conclusion

In this work, we presented a fast, instance-optimized slice-to-volume reconstruction network for the generation of 3D cine MRI from free-breathing multi-planar 2D acquisitions. Validated on simulated phantom MRI and in-vivo real-time MRI, the model demonstrated potential to derive isotropic, spatiotemporally consistent 3D+t representations from anisotropic 2D cine MRI. Using motion states extracted from ECG and respiration belt sensors, it optimized cardiac and respiratory motion fields, achieving promising inter-slice movement compensation. Further improvements are needed to enhance reconstruction sharpness and to refine learnt cardiac contraction patterns. Extensions of the network will be explored to address irregular rhythm disorders, leveraging advanced arrhythmia motion-state encodings and fine-tuning strategies to account for motion state imbalances.

6 Supplementary information

The supplemental material provides implementation details and videos of example reconstructions. The videos show free-breathing phantom reconstructions (SI Fig.1a and Fig.1b), a breath-hold phantom reconstruction (SI Fig.2), and in-vivo reconstructions SI Fig.3–5.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s11517-025-03488-7>.

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Declarations

Competing Interests The authors declare no conflicts of interest.

Ethics Approval and Consent The data was acquired under the approved ethical protocol “METHODO” (ClinicalTrials.gov Identifier: NCT02887053). The study was approved by the institutional ethics review board (CPP EST-III, 08.10.01). All participants provided informed written consent.

Consent for Publication The source code will be made publicly available. Restrictions apply to the availability of XCAT data presented in the referenced work of Segars et al. The XCAT phantoms were generated with the permission of the Duke University (4D Extended Cardiac-Torso (XCAT) Phantom Version 2.0), DUKE UNIVERSITY, Durham, NC 27708. The in-vivo data that support the findings of this study are not openly available due to reasons of sensitivity. Data are located in controlled access data storage at the University Hospital in Nancy.

Generative AI in Scientific Writing During the preparation of this work the first author used GPT-4 to improve the grammar and readability of certain paragraphs. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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